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Education

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
M.Tech. (Robotics)	Indian Institute of Technology, Jodhpur	7.68 (Current)	2025-Present
B.Tech. (ECE)	Assam University, Silchar	7.61	2020-2024
Senior Secondary	CBSE Board	78%	2020
Secondary	CBSE Board	9.8	2017

Experience

- **Skillonation Edtech** *July 2026 - Aug. 2026*
  Robotics and Embedded Intern Jodhpur
  - Designed and built an automated **notebook page-turning and scanning machine** from scratch, targeting 50 pages/minute throughput.
  - Engineered full hardware stack: ESP32-based control system, conveyor belt with proximity sensor-triggered stops, Bernoulli gripper on pneumatic actuator, stepper-driven page-turning knife, and solenoid valve switching for pneumatic control.
  - Integrated dual IMX477 CSI cameras on Jetson Nano with UART-triggered capture pipeline using OpenCV for high-quality page imaging.
  - Interfaced industrial components including Janatics proportional pressure regulator, SMC ZK2 vacuum ejector, and CEYONE photoelectric sensors with ESP32 via relay and optocoupler level-shifting circuits.

Projects

- **Robotic Disassembly of Electronic Devices using HIL-SERL** *May 2026 - Present*
  M.Tech. Thesis | Advisor: Dr. Roshan Kumar Hota, IIT Jodhpur Github
  - Implementing **Hardware-in-the-Loop Sample Efficient Reinforcement Learning (HIL-SERL)** for autonomous robotic disassembly of consumer electronic devices using an **AgileX Piper X** 6-DOF manipulator arm.
  - Built full hardware-software stack: RealSense D435i wrist camera, overhead RGB camera, gamepad-based teleoperation for human demonstration collection, and reward classifier trained on collected data.
  - Completed teleoperation pipeline, demonstration data collection, and reward classifier training; online RL training (SAC/RLPD) in progress for pick-and-place disassembly primitives.
  - **Tools & Technologies**: ROS2, Python, PyTorch, SAC/RLPD, OpenCV, RealSense SDK, AgileX Piper SDK

- **Bimanual Manipulation via Fine-tuned Vision-Language-Action Model** *Dec. 2025 - Apr. 2026*
  SmolVLA | HuggingFace LeRobot | SO-101 Dual-Arm Platform Github | Demo Video
  - Fine-tuned **SmolVLA** (HuggingFace) on a **bimanual coordination task**: right arm passes a red bowl to the left arm, which then picks up a box and places it into a green bowl — requiring cross-arm temporal dependency.
  - Collected teleoperation demonstrations on dual **SO-101** robotic arms using LeRobot framework; trained end-to-end visuomotor policy from RGB observations to joint-space actions.
  - Deployed trained VLA policy on physical hardware achieving consistent task completion, validating real-world transfer of learned bimanual coordination.
  - **Tools & Technologies**: SmolVLA, LeRobot, Python, PyTorch, SO-101, HuggingFace Transformers

- **Autonomous Mobile Robot — Full Stack Development** *Jan. 2025 - May 2025*
  Custom Differential Drive | ROS2 | Nav2 | SLAM Toolbox | Raspberry Pi Github
  - **CAD Design & Robot Visualization**: Designed complete custom differential-drive robot chassis from scratch in SolidWorks; converted to URDF/XACRO for ROS2 and visualized joint states, TF frames, and sensor placement in RViz2.
  - **Joystick Teleoperation**: Implemented joystick-based teleoperation using `teleop_twist_joy` and custom ROS2 nodes publishing to `/cmd_vel`; validated motor response and odometry feedback on physical hardware.
  - **Autonomous Maze Solver**: Deployed reactive navigation algorithm using RPLidar scan data to detect walls and navigate through maze environments without prior map knowledge.
  - **SLAM Mapping**: Integrated **SLAM Toolbox** with RPLidar A-series for real-time 2D occupancy grid mapping; tuned scan-matching parameters and loop closure for accurate large-environment maps.
  - **Autonomous Navigation**: Configured **Nav2** stack with AMCL localization on saved SLAM maps; tuned DWB local planner and costmap inflation radius for collision-free point-to-point navigation.
  - **Autonomous Docking**: Implemented ArUco marker-based docking using onboard camera; ROS2 node estimates marker pose via `cv2.aruco`, computes alignment error, and issues corrective commands for precise dock approach.

– **Hardware Deployment**: Deployed full ROS2 stack on **Raspberry Pi** onboard compute; configured systemd launch services, tuned RPLidar driver parameters, and validated complete navigation pipeline on physical robot in real-world environment.

- **VLA for Autonomous Driving — Code as Policy** *Oct. 2026 - Present*
  SmolVLA | ROS2 | CARLA Simulator | Claude Code Agent Github
  – Implementing a **Code-as-Policy** framework for autonomous driving inspired by the CAPX (NVIDIA, 2026) paper, using a coding agent as a zero-shot visuomotor controller in place of a pre-trained VLA model.
  – Designing a lightweight 3D simulation environment with top-down, chase, and front camera APIs; coding agent calibrates environment physics and issues motor commands (vx, vy, ω) solely via GET/POST API calls without direct access to simulation code.
  – Evaluating agent performance under goal randomization and dynamic obstacle perturbations; planned extension to CARLA simulator for high-fidelity autonomous driving benchmarking.

- **World Models for Robotic Manipulation — From Scratch** *Nov. 2026 - Present*
  Dreamer | IRIS | DINO-WM | PyTorch | SO-101 Github
  – Implementing world models from scratch following the simulator-renderer-planner taxonomy; building Mini-Dreamer (continuous latent simulator with reconstruction training) and Mini-IRIS (token-space video game agent) as foundational builds.
  – Studying latent-space prediction approaches (JEPA-style, no pixel reconstruction) vs. pixel-space approaches (IRIS) to understand trade-offs in representation, training signal, and inference latency for robotic deployment.
  – Target application: deploying a learned world model on **SO-101** robotic arm to enable sample-efficient planning in imagination without real-world rollouts, complementing ongoing HIL-SERL disassembly research.

Key Courses Taken

- Machine Learning, Deep Learning, Introduction to Robotics, Mobile Robots, Advanced Robotic Manipulation, Computer Vision, Robot Learning

Technical Skills

- **Programming:** Python, C/C++, MATLAB, Arduino (ESP32)
- **Robot Frameworks:** ROS2 Humble, LeRobot, PyAgxArm SDK, Piper SDK
- **ML/RL Libraries:** PyTorch, Stable-Baselines3, HuggingFace Transformers, OpenCV
- **Simulation Tools:** MuJoCo, Gazebo , RViz, Isaac Sim, Tinkercad
- **Hardware & Embedded:** AgileX Piper X, SO-101, ESP32, Jetson Nano/Orin, RealSense D435i, IMX477 CSI Camera, Feetech STS3215
- **Pneumatics & Actuation:** Janatics PPR, SMC ZK2 Vacuum Ejector, Solenoid Valves, Bernoulli Gripper, Stepper Drivers (Leadshine CS-D508)
- **Tools & Platforms:** Git, SolidWorks, Linux (Ubuntu), Docker, Overleaf

Achievements

- **Runner-Up** Smart India Hackathon (SIH 2023), MHRD, Government of India. Developed OCR-based automated attendance

Certifications

- Modern Robot Learning — Vizura AI
- ROS2 MoveIt2 - Control a Robotic Arm — Udemy
- ROS2 for Beginners Level 2 - TF | URDF | RViz | Gazebo — Udemy
- Arduino Programming with Tinkercad — Udemy

Positions of Responsibility

- **Department Representative — ECE** Assam University, Silchar *Aug. 2022 - Aug. 2023*